EE2021 -
Materials Science for Electrical Engineers

L3: Properties of Materials

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Materials Science vs Engineering

- Materials science involves investigating the relationships that exist between the structures and properties of materials (i.e., why materials have their properties)

- Materials engineering involves, on the basis of these structure–property correlations, designing or engineering the structure of a material to produce a predetermined set of properties

Categories of materials

- Metals
- Ceramics (compounds between metallic and nonmetallic elements)
- Glasses (Amorphous transparent solids)
- Polymers (compounds composed of carbon, hydrogen, and other nonmetallic elements)
- Composites (composed of at least two different material types)
- Semiconductors

Property

- A measurement is made of the response of a material, in terms of a state parameter (e.g. strain, change in temperature or current density), to the influence of an external effect or field parameter (e.g. stress, amount of heat input or electric field strength). The quotient (could be a tensor) of these two measurements is then the material property (e.g. elastic modulus, specific heat capacity or electrical conductivity).

- In order to measure the properties of a material we do not need to know anything about its internal structure.

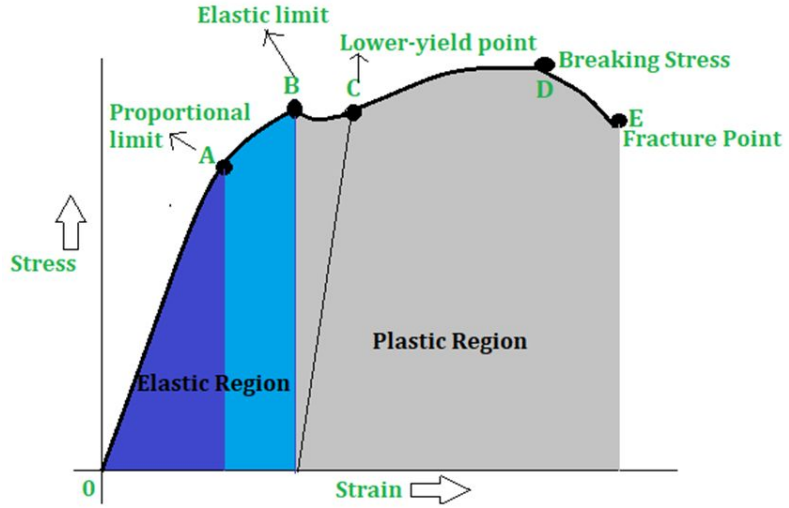
Macroscopic (Bulk) Properties of Materials (Response and Stimulus)

- Electrical (Conductivity/permittivity vs Electric field)
- Thermal (Change in temp/gradient or expansion vs heat)
- Magnetic (Susceptibility vs magnetic field strength)
- Mechanical (deformation vs applied force/stress, resistance to fracture, hardness to scratch)
- Optical (refracted vs incident angle, attenuation of intensity)
- Chemical or Deteriorative characteristics (Chemical reactivity, Corrosion rate)

They could be also coupled across domains - Piezoelectricity, piezoresistance, magnetoresistance, thermoelectricity, photoconductivity etc

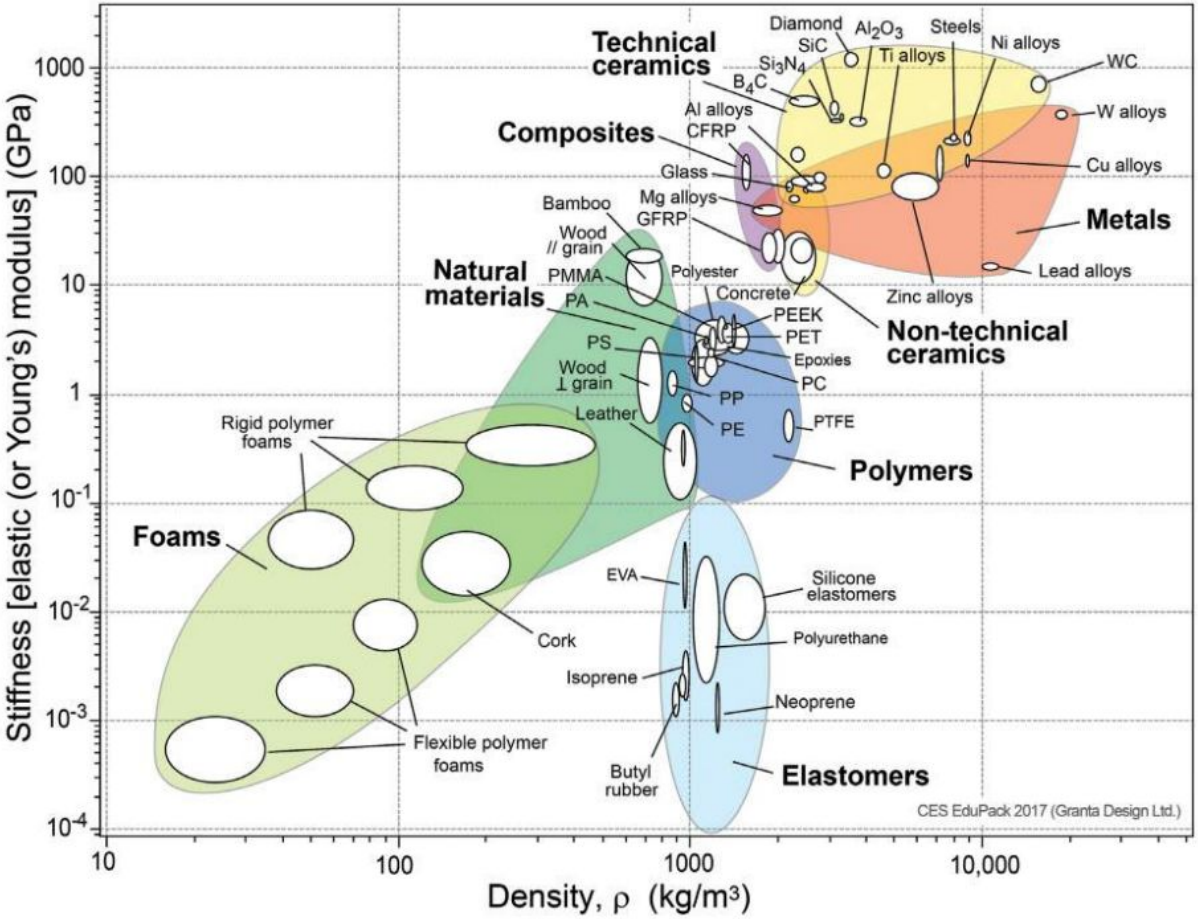
Mechanical Properties

- Modulus of elasticity
- Yield strength
- Tensile strength
- Ductility
- Resilience
- Fracture Toughness
- Hardness
- Fatigue
- Creep



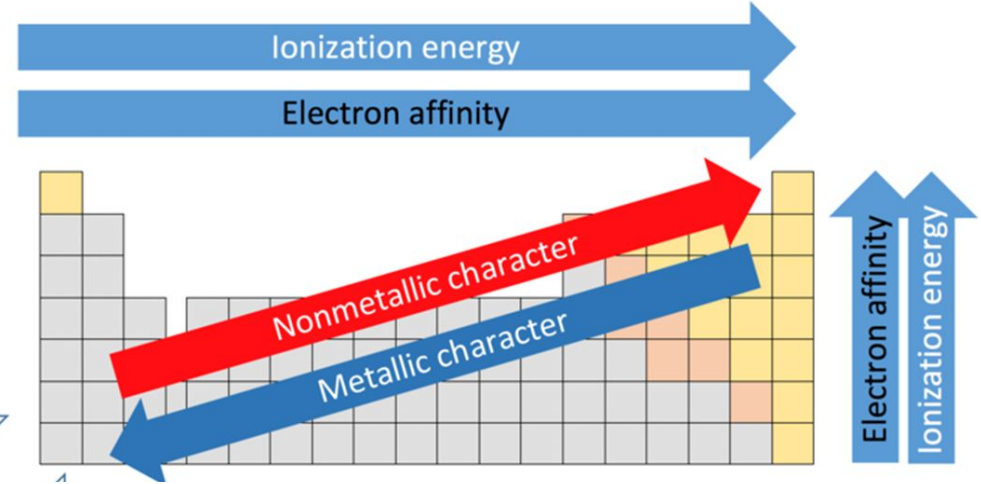


Mechanical Properties of different Materials



Chemical Properties

- Bonding Energy
- Ionization energy
- Electron affinity
- Deteriorative properties

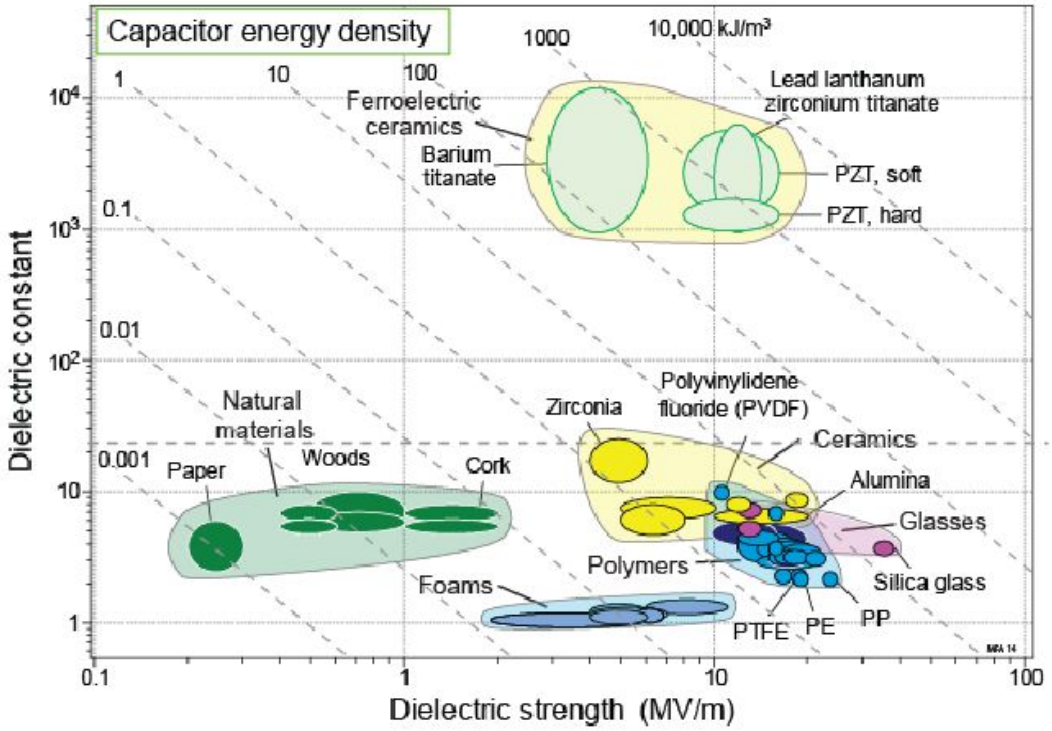


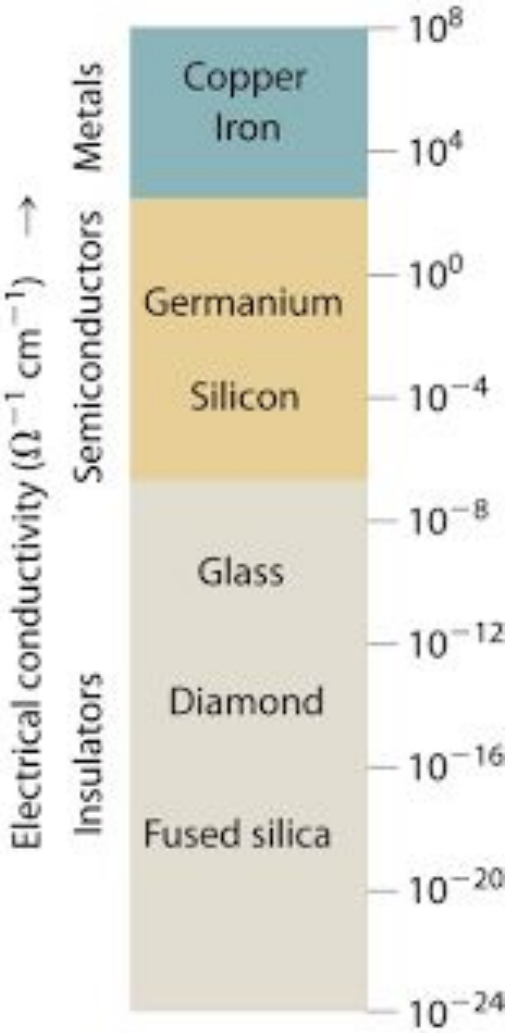


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Electrical Properties

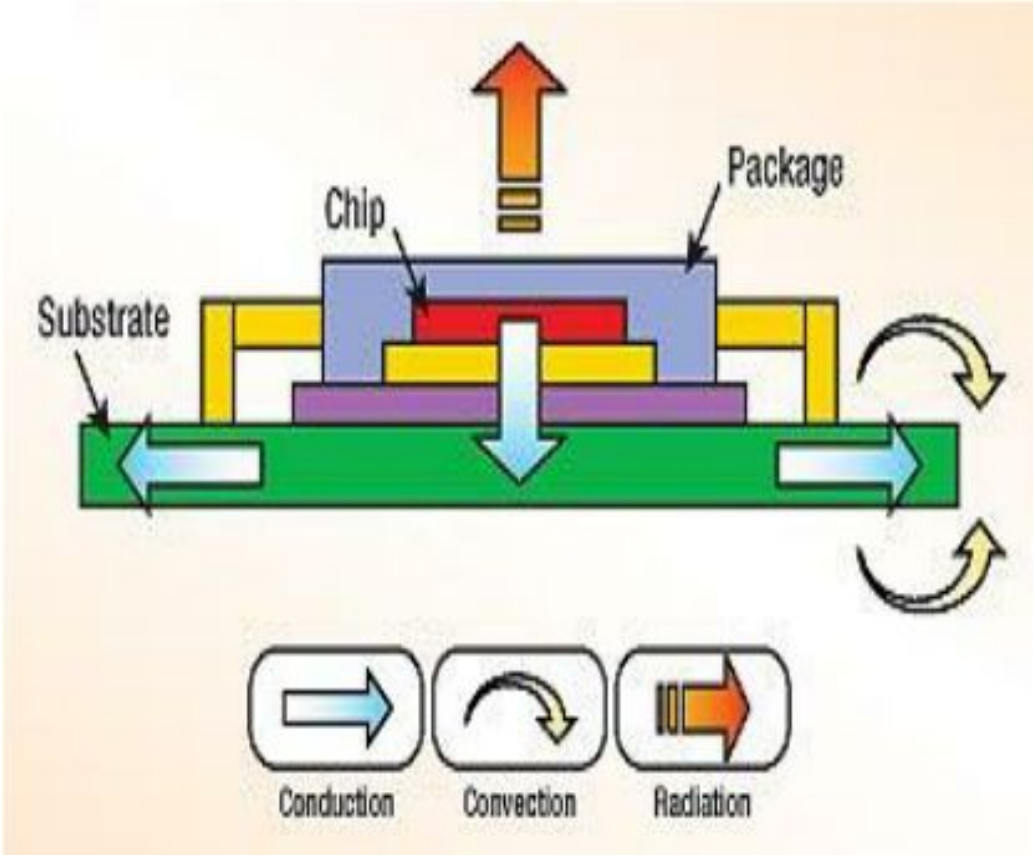
- Electrical conductivity
- Dielectric constant





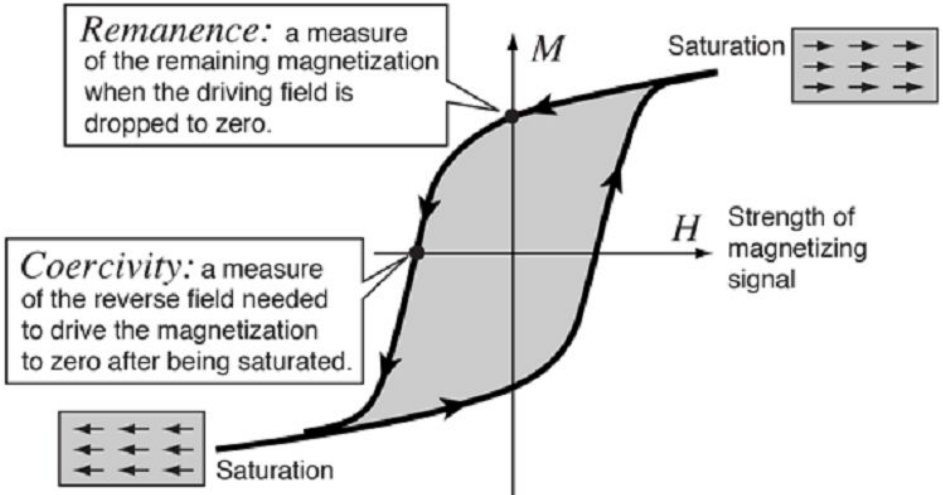
Thermal Properties

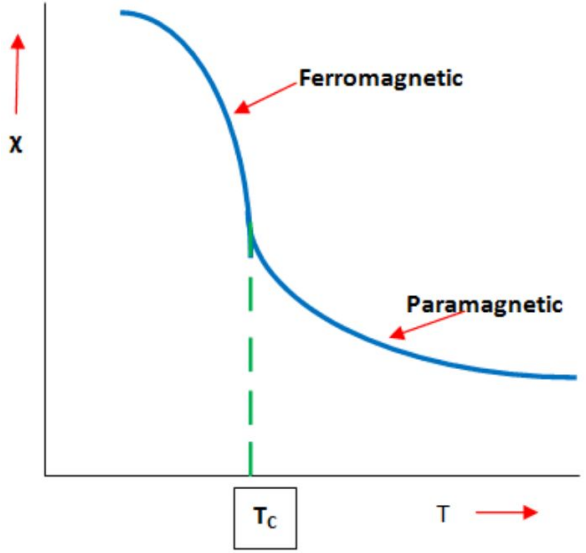
- Thermal expansion
- Thermal conductivity
- Specific heat

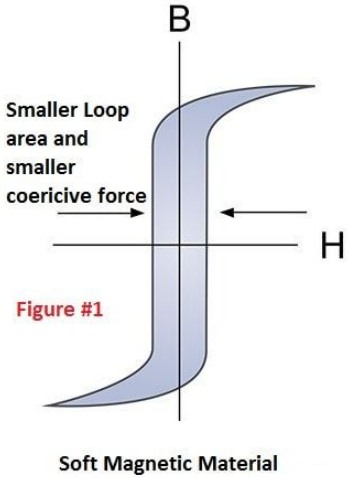


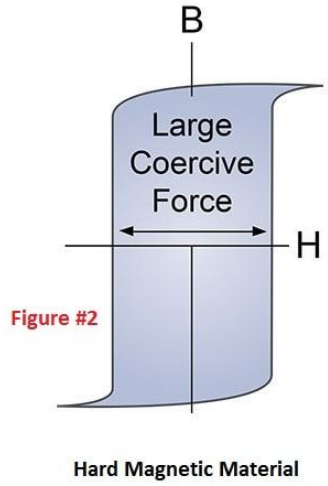
Magnetic Properties

- Curie temperature
- Remanence
- Coercivity
- Soft Magnetic materials
- Hard Magnetic materials



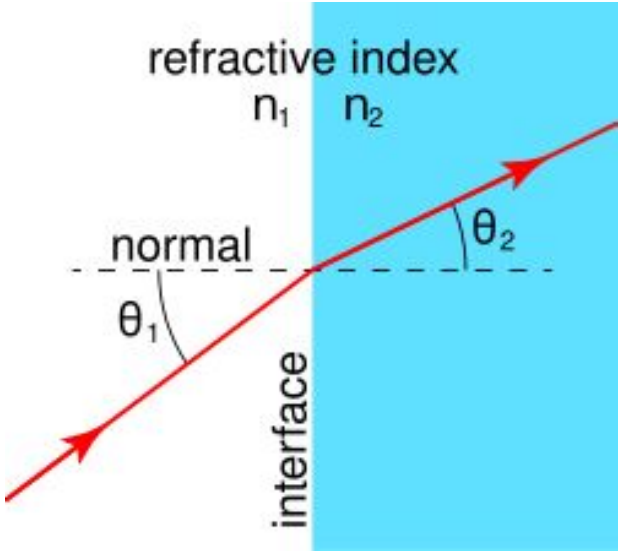


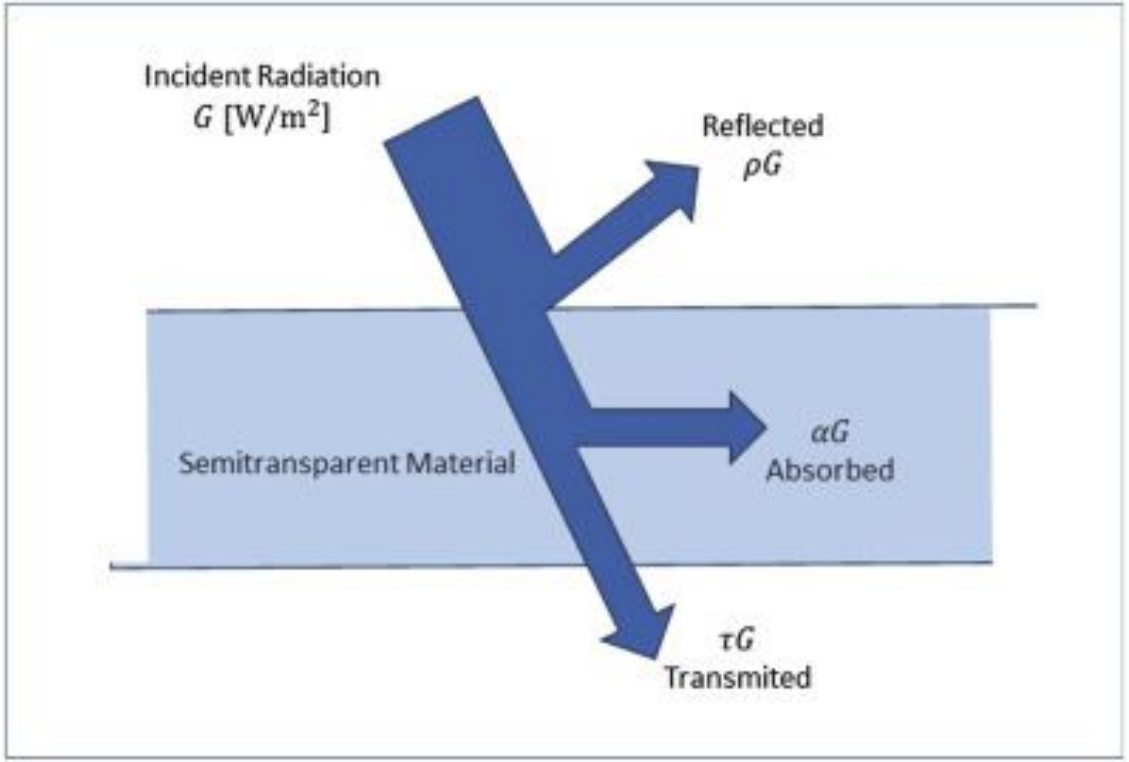




Optical Properties

- Refractive index
- Reflectivity
- Absorptivity
- Transmissivity





Difficult questions following easy (?) measurements

- Why does a material display a specific value for a specific property from the range of values possible for that property?
- What are the limits, if any, to this property?
- Does the value of a property remains invariant with external conditions?
- Why are some properties correlated? Electrical and thermal conductivity, optical reflectivity and electrical conductivity, Chemical etch rate and hardness
- Is there a more fundamental explanation (which can also make bold predictions*) involving micro/atomic/electronic structure?

*What do heavens tell us about modeling?

Correlated properties and trends

- The Hagen-Rubens law (optical reflectivity and electrical conductivity at long wavelengths > 3 μm)

$$R = 1 - 2\sqrt{\frac{4\pi\varepsilon_0\nu}{\sigma_0}}$$

- The Wiedemann-Franz law

Table 3.2 Lorenz numbers $\kappa/(T\sigma)$ for various metals in units of 10^{-8} WattOhm/K^2

Material	L
Lithium (Li)	2.22
Sodium (Na)	2.12
Copper (Cu)	2.20
Iron (Fe)	2.61
Bismuth (Bi)	3.53
Magnesium (Mg)	2.14

Table 2.1 Heat capacities of some solids at room temperature and pressure.

Material	C/R
Aluminum (Al)	2.91
Antimony (Sb)	3.03
Copper (Cu)	2.94
Gold (Au)	3.05
Silver (Ag)	2.99
Diamond (C)	0.735

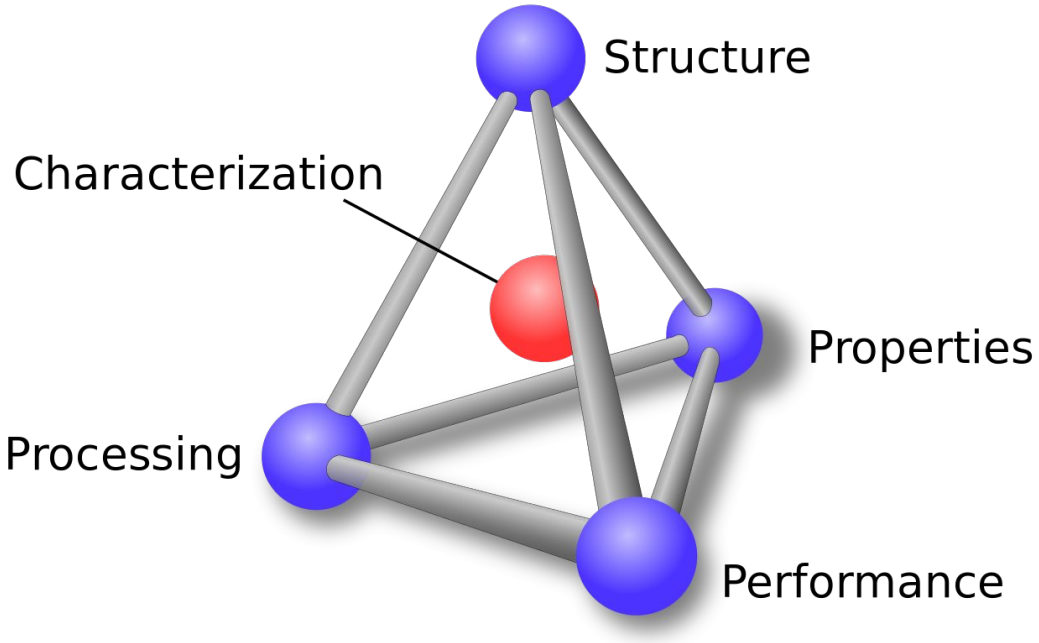
- Dulong-Petit's law

Structure of Material

Structure of a material usually relates to the arrangement of its internal components

- **Electronic structure** refers to the arrangement and behavior of electrons within material, including their distribution among energy levels and spatial locations.
- **Atomic structure**—relates to the organization of atoms to yield molecules or crystals.
- **Nanostructure**—deals with aggregates of atoms that form particles (nanoparticles) that have nanoscale dimensions (< 100 nm)
- **Microstructure**—those structural elements that are subject to direct observation using some type of microscope (> 100 nm & < mm).
- **Macrostructure**—structural elements that may be viewed with the naked eye (with scale range between several mm to ~ m).

Materials Tetrahedron



Electron microscopy and x-ray analysis are perhaps the two most frequently used characterization techniques to “see/infer” the micro/atomic/electronic structure

Materials selection

The more familiar an engineer or scientist is with the various characteristics and structure–property relationships, as well as the processing techniques of materials, the more proficient and confident he or she will be in making judicious materials choices.

Four important criteria in materials selection

- in-service conditions to which the material will be subjected
- any deterioration of material properties during operation
- economics or cost of the fabricated piece
- Toxicity or harmful nature of materials

Summary

- Several properties of materials are related to each other
- Important to develop a model that helps us understand the origin of these properties and the manner in which the properties relate to each other